Note: ECO2010 Problem Set 4, Question 2

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Ω UHC Argument

We want to show that $\Omega(\boldsymbol{p}, b)$ is UHC. Suffices to show that, given $(\boldsymbol{p}_0, b_0)$, with $b_0 > 0$ and $\boldsymbol{p}_{0,(i)} > 0\ \forall i$, and V open such that $\Omega(\boldsymbol{p}_0, b_0) \subseteq V$, there exists $\delta > 0$ such that $\Omega(\boldsymbol{p}, b) \subseteq V$ for all $(\boldsymbol{p}, b) \in N_\delta(\boldsymbol{p}_0, b_0)$, because then the open set "U" needed to satisfy the UHC requirement can just be defined as $U := N_\delta(\boldsymbol{p}_0, b_0)$.

For convenience, we will use the max metric throughout, i.e.

$$d(\boldsymbol{x}, \boldsymbol{y}) := \|\boldsymbol{x} - \boldsymbol{y}\|_\infty := \max_i |\boldsymbol{x}_{(i)} - \boldsymbol{y}_{(i)}| .$$

Since V is open and $\Omega(\boldsymbol{p}_0, b_0) \subseteq V$, we have that, for all $\boldsymbol{x} \in \Omega(\boldsymbol{p}_0, b_0)$, there exists $\epsilon_{\boldsymbol{x}} > 0$ such that $N_{\epsilon_{\boldsymbol{x}}}(\boldsymbol{x}) \subseteq V$. Now, $\left\{N_{\epsilon_{\boldsymbol{x}}/2}(\boldsymbol{x}) \,\middle|\, \boldsymbol{x} \in \Omega(\boldsymbol{p}_0, b_0)\right\}$ is an open cover of $\Omega(\boldsymbol{p}_0, b_0)$ and therefore, since $\Omega(\boldsymbol{p}_0, b_0)$ is compact, this open cover has a finite subcovering, i.e. there exists a finite set $\{\boldsymbol{x}_1, \ldots \boldsymbol{x}_K\} \subseteq \Omega(\boldsymbol{p}_0, b_0)$ such that $\Omega(\boldsymbol{p}_0, b_0) \subseteq \bigcup_{k=1}^K N_{\epsilon_{\boldsymbol{x}_k}/2}(\boldsymbol{x}_k) \subseteq V$. Define

$$\epsilon := \min_{k \in \{1, \ldots, K\}} \frac{\epsilon_{\boldsymbol{x}_k}}{2} > 0 .$$

Now, it suffices to show that there exists $\delta > 0$ such that, for all $(\boldsymbol{x}, \boldsymbol{p}, b)$ such that $(\boldsymbol{p}, b) \in N_\delta(\boldsymbol{p}_0, b_0)$ and $\boldsymbol{x} \in \Omega(\boldsymbol{p}, b)$, there exists $\boldsymbol{x}^\dagger \in \Omega(\boldsymbol{p}_0, b_0)$ such that $\boldsymbol{x} \in N_\epsilon\left(\boldsymbol{x}^\dagger\right)$, since then $\boldsymbol{x} \in N_{\epsilon_{\boldsymbol{x}_k}}(\boldsymbol{x}_k)$ for some $\boldsymbol{x}_k$, and so, $\boldsymbol{x} \in V$.

Now, define $\underline{p} := \min_i \boldsymbol{p}_{0,(i)} > 0$, and note that, for sufficiently small $\delta > 0$, we must have $\boldsymbol{p}_{\delta,(i)} \geq \underline{p}/2$, and so, we have

$$\boldsymbol{x}_{\delta,(i)} \leq \frac{b_\delta}{\boldsymbol{p}_{\delta,(i)}} \leq \frac{2b_\delta}{\underline{p}} < \frac{2(b_0 + \delta)}{\underline{p}} ,$$

which implies

$$\begin{aligned}\boldsymbol{p}_0 \cdot \boldsymbol{x}_\delta = \boldsymbol{p}_\delta \cdot \boldsymbol{x}_\delta + (\boldsymbol{p}_0 - \boldsymbol{p}_\delta) \cdot \boldsymbol{x}_\delta \leq b_\delta + n\delta \|\boldsymbol{x}_\delta\|_\infty \leq b_\delta + n\delta\left(\frac{2(b_0+\delta)}{\underline{p}}\right) \\ \leq b_0 + \delta + n\delta\left(\frac{2(b_0+\delta)}{\underline{p}}\right) .\end{aligned}$$

Now, we have that

$$\boldsymbol{p}_0 \cdot (\lambda \boldsymbol{x}_\delta) \leq \lambda \left(b_0 + \delta + n\delta \left(\frac{2\,(b_0 + \delta)}{\underline{p}} \right) \right) \ ,$$

and so, setting $\boldsymbol{x}_\delta^\dagger := \lambda_\delta^\dagger \boldsymbol{x}_\delta$, where

$$\lambda_\delta^\dagger := \frac{b_0}{b_0 + \delta + n\delta \left(\frac{2(b_0+\delta)}{\underline{p}} \right)} \in (0,1) \ ,$$

and we have that

$$\boldsymbol{p}_0 \cdot \boldsymbol{x}_\delta^\dagger \leq b_0 \ .$$

Finally, note that $\lambda_\delta^\dagger$ approaches 1 as δ approaches 0, and so,

$$\|\boldsymbol{x}_\delta - \boldsymbol{x}_\delta^\dagger\|_\infty = \left\| \left(1 - \lambda_\delta^\dagger\right) \boldsymbol{x}_\delta \right\|_\infty \leq \left(1 - \lambda_\delta^\dagger\right) \frac{2\,(b_0 + \delta)}{\underline{p}}$$

can be made arbitrarily small by picking δ sufficiently small. The result now follows immediately by picking $\delta > 0$ small enough that

$$\left(1 - \lambda_\delta^\dagger\right) \frac{2\,(b_0 + \delta)}{\underline{p}} < \epsilon \ .$$